

XIAOLEI (VINCENT) HU

AUTOMATION, CONTROLS & ROBOTICS ENGINEER

San Francisco, CA | +1 917-436-0680 | vincent.hu@hucrafts.com | [linkedin.com/in/xiaoleih](https://www.linkedin.com/in/xiaoleih)

Professional Summary

Automation and controls engineer with 6+ years of experience delivering PLC-controlled machines, precision motion systems, industrial HMIs, Python services, and electromechanical products. Combines mechanical design, TwinCAT Structured Text, Ignition, sensor integration, and data-driven optimization to build maintainable, production-ready automation systems.

Professional Experience

Mainspring Energy

2026

Senior Controls & Automation Engineer

- Engineered a closed-loop power-cylinder alignment platform in TwinCAT 3 Structured Text, coordinating 14+ program blocks for multi-axis servo motion, sensor acquisition, calibration, sweep sequences, and iterative correction; reduced the overall alignment cycle time by 75%, from 8 hours to 2 hours.
- Achieved 5-micron two-position coaxial alignment using Jacobian-based inverse kinematics and 50-micron ten-position coaxial optimization using SVD-based 3D regression with cost-function minimization.
- Built an 88-step Ignition Vision workflow with dynamic navigation, pass/fail routing, state tracking, database logging, and OPC UA connectivity to the PLC.
- Integrated EtherCAT, ADS, OPC UA, RS-422, and TCP/IP communications into a cohesive real-time controls architecture running on a Beckhoff IPC.

Myriad Genetics

2022 - 2026

Hardware Automation Engineer

- Delivered three automated, high-throughput genetic screening workcells, coordinating cross-functional teams through instrument qualification, system integration testing, and regulatory-driven design.
- Architected the SkySlide inter-laboratory sample-conveyance system, integrating mechanical design, PLC motion control, power distribution, optical sensing, and site infrastructure to increase automation usage from 34% to 80%.
- Engineered collision-free multi-conveyor state machines with recoverable motion logic and plate alignment and presence detection using optical sensor data.
- Modernized engineering data infrastructure through Arena PLM and Onshape migrations; automated 3,000+ component records using Jupyter, Pandas, and VBA, reducing processing time from weeks to minutes.
- Built a Django- and Git-based version control system for robotic liquid-handler methods, improving protocol traceability and automating change-control workflows.

Professional Experience, Continued

Kinnos, Inc. <i>Mechanical Engineer</i>	2020 - 2022
• Delivered two generations of electromechanical disinfection devices, leading mechanical design, supplier coordination, production readiness, and QMS documentation for injection-molded and assembled components. • Maintained production documentation across CAD, manufacturing process instructions, inspections, material review board workflows, and quality records. • Engineered Python- and C#-based end-of-line inspection and reliability fixtures for firmware flashing, pump calibration, and database logging, reducing inspection time by 79% and 27%.	

Technical Skills

- **Controls & HMI:** TwinCAT 3, IEC 61131-3 Structured Text, TwinCAT MC2, Ignition Vision, OPC UA, ADS, EtherCAT, RS-422, Modbus TCP, TCP/IP
- **Programming & Data:** Python, FastAPI, NumPy, Pandas, C#, Django, REST APIs, Jupyter, Git, shell scripting
- **Mechanical & PLM:** SOLIDWORKS, Onshape, GD&T, DFM/DFA, electromechanical design, Arena PLM, ECO and BOM workflows
- **Robotics & Simulation:** ROS 2, RViz, Gazebo, CoppeliaSim, Lua, motion control, state machines
- **Languages:** English (fluent), Mandarin Chinese (fluent), Japanese (basic)

Education

Johns Hopkins University <i>M.S. Robotics & Autonomous Systems</i>	2024 - Present GPA: 4.0/4.0
The Cooper Union for the Advancement of Science and Art <i>B.E. Mechanical Engineering</i>	2016 - 2020 GPA: 3.6/4.0
• Tau Beta Pi, The Engineering Honor Society; Pi Tau Sigma, National Mechanical Engineering Honor Society.	

Certifications

- Certified SOLIDWORKS Expert (CSWE).
- Certified SOLIDWORKS Professional Advanced: Surfacing, Weldments, Sheet Metal, and Drawing Tools.
- TwinCAT 3 PLC Training, Beckhoff Automation.
- Onshape Bootcamp, Onshape.